

Beta Funktion

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)} \quad m > 0, n > 0$$

a) Evaluate

$$\int_0^1 x^4 (1-5x)^5 dx$$

$$\therefore \beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$\begin{aligned} u &= 5x, \quad x = u^2 \\ du &= \frac{1}{2\sqrt{u}} du \\ dx &= \frac{1}{2\sqrt{u}} du \end{aligned}$$

$$\int_0^1 (u^2)^4 \cdot (1-u)^5 \cdot 2u du$$

$$2 \int_0^1 u^{10-1} (1-u)^{6-1} du$$

$$m=10, \quad n=6$$

$$= \frac{\Gamma(10) \Gamma(6)}{\Gamma(16)}$$

$$= \frac{9! \cdot 5!}{15!}$$

$$= \frac{1}{15015}$$

$$b) \int_0^1 x^2 (1-x^2)^2 dx$$

$$\begin{aligned} u &= x^2, \quad x = u^{\frac{1}{2}} \\ du &= 2x dx \\ dx &= \frac{1}{2x} du \\ \text{or} \\ dx &= \frac{1}{2u^{\frac{1}{2}}} du \end{aligned}$$

$$\int_0^1 (u^{\frac{1}{2}})^2 \cdot (1-u)^2 \cdot \frac{1}{2u^{\frac{1}{2}}} du$$

$$\frac{1}{2} \int_0^1 \frac{u^1}{u^{\frac{1}{2}}} \cdot (1-u)^2 du$$

$$\frac{1}{2} \int_0^1 u^{\frac{1}{2}-1} \cdot (1-u)^{3-1} du$$

$$\frac{1}{2} \beta\left(\frac{3}{2}, 4\right)$$

$$= \frac{\Gamma\left(\frac{3}{2}\right) \times \Gamma(4)}{\Gamma\left(\frac{3}{2}+4\right)}$$

$$= \frac{1}{2} \cdot \frac{\frac{1}{2}\sqrt{\pi} \times 3!}{\Gamma\left(\frac{11}{2}\right)}$$

$$\frac{1}{2} \cdot \frac{\frac{1}{2}\sqrt{\pi} \times 6}{\frac{945}{32}\sqrt{\pi}}$$

$$= \frac{16}{315}$$

$$\therefore \frac{2}{3} \times \sqrt{\frac{2}{\pi}}$$

$$\frac{1}{2} \times \frac{2}{2} \cdot \sqrt{\frac{2}{\pi}}$$

$$\frac{1}{2} \times \frac{2}{2} \times \frac{2}{2} \times \frac{2}{2} \times \frac{1}{2} \sqrt{\frac{2}{\pi}}$$

$$\frac{945}{32} \sqrt{\pi}$$

$$e) \int_0^1 \sqrt{t} (1-t)^{\frac{1}{2}} dt$$

$$\int_0^1 t^{\frac{1}{2}-1} (1-t)^{\frac{1}{2}-1} dt$$

$$B\left(\frac{1}{2}, \frac{1}{2}\right)$$

$$= \frac{\sqrt{\frac{1}{2}} \times \sqrt{\frac{1}{2}}}{\Gamma\left(\frac{1}{2}\right)}$$

$$= \frac{\frac{1}{2}\sqrt{\pi} \times \frac{1}{2}\sqrt{\pi}}{2}$$

$$= \frac{1}{8} \pi$$

$$e) \int_0^1 x^3 (1-x)^3 dx$$

